DIMENSION OF A FIBRE OVER A ONE-DIMENSIONAL LOCAL RING FROM THE DIMENSION AT GENERIC POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the two matching inequalities; no step is referred to the reader.

Editorial note. The counted spans carry 5 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-dimension-fibre-in-codim-1 is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Spec of the local ring at y is the set of points specializing to y. Schemes, Lemma schemes-lemma-specialize-points, invoked at proof lines 3606-3608. Supplies the reduction of the base to the local ring at y.
- Dimension formula for a morphism of schemes. Morphisms, Lemma morphisms-lemma-dimension-formula, invoked at proof lines 3626-3627. Supplies the inequality from which the upper bound on the fibre dimension is read off.
- Dimension of a fibre at a point as a sum of the local ring dimension and a transcendence degree. Morphisms, Lemma morphisms-lemma-dimension-fibre-at-a-point, invoked at proof lines 3635-3636 and 3642-3646. Supplies the identification of the fibre dimension with the two expressions compared by the dimension formula.
- Dimension of fibres is bounded below by the dimension at the generic point of the component. Morphisms, Lemma morphisms-lemma-openness-bounded-dimension-fibres, invoked at proof lines 3647-3649. Supplies the reverse inequality dim_x(X_y) >= d.

Source reference. Original lemma 02FX: sources/stacks-chow/tags.txt, source lines 3208–3208. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02JU: sources/stacks-chow/tags.txt, source lines 3349–3349. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 02FZ: sources/stacks-chow/tags.txt, source lines 3210–3210. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01J7: sources/stacks-chow/tags.txt, source lines 2004–2004. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to Y$ be locally of finite type. Let $x \in X$ be a point with image $y \in Y$ such that $\mathcal{O}_{Y,y}$ is Noetherian of dimension ≤ 1. Let $d \geq 0$ be an integer such that for every generic point η of an irreducible component of X which contains x, we have $\dim_\eta(X_{f(\eta)}) = d$. Then $\dim_x(X_y) = d$.*

Proof. Recall that $\mathrm{Spec}(\mathcal{O}_{Y,y})$ is the set of points of Y specializing to y, see Schemes, Lemma 01J7. Thus we may replace Y by $\mathrm{Spec}(\mathcal{O}_{Y,y})$ and assume $Y = \mathrm{Spec}(B)$ where B is a Noetherian local ring of dimension ≤ 1 and y is the closed point. We may also replace X by an affine neighbourhood of x.

Let $X = \bigcup X_i$ be the irreducible components of X viewed as reduced closed subschemes. If we can show each fibre $X_{i,y}$ has dimension d, then $X_y = \bigcup X_{i,y}$ has dimension d as well. Thus we may assume X is an integral scheme.

If $X \to Y$ maps the generic point η of X to y, then X is a scheme over $\kappa(y)$ and the result is true by assumption. Assume that X maps η to a point $\xi \in Y$ corresponding to a minimal prime $\mathfrak{q}$ of B different from $\mathfrak{m}_B$. We obtain a factorization $X \to \mathrm{Spec}(B/\mathfrak{q}) \to \mathrm{Spec}(B)$. By the dimension formula (Morphisms, Lemma 02JU) we have

$$\dim(\mathcal{O}_{X,x}) + \mathrm{trdeg}_{\kappa(y)}\kappa(x) \leq \dim(B/\mathfrak{q}) + \mathrm{trdeg}_{\kappa(\mathfrak{q})}(R(X))$$

We have $\dim(B/\mathfrak{q}) = 1$. We have $\mathrm{trdeg}_{\kappa(\mathfrak{q})}(R(X)) = d$ by our assumption that $\dim_\eta(X_\xi) = d$, see Morphisms, Lemma 02FX. Since $\mathcal{O}_{X,x} \to \mathcal{O}_{X_s,x}$ has a kernel (as $\eta \mapsto \xi \neq y$) and since $\mathcal{O}_{X,x}$ is a Noetherian domain we see that $\dim(\mathcal{O}_{X,x}) > \dim(\mathcal{O}_{X_y,x})$. We conclude that

$$\dim_x(X_s) = \dim(\mathcal{O}_{X_s,x}) + \mathrm{trdeg}_{\kappa(y)}\kappa(x) \leq d$$

(Morphisms, Lemma 02FX). On the other hand, we have $\dim_x(X_s) \geq \dim_\eta(X_{f(\eta)}) = d$ by Morphisms, Lemma 02FZ. □